National Science Foundation
Office of Science and Technology Policy
Networking and Information Technology
Research and Development National Coordination Office

Response to RFI on the Development of an Artificial Intelligence (AI) Action Plan

March 14, 2025

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Mr. Faisal D'Souza
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue
Alexandria, VA 22314

RE: Deloitte Consulting, LLP Comments on the "Development of an Artificial Intelligence (AI) Action Plan"

Dear Mr. D'Souza:

Deloitte Consulting LLP (Deloitte[1]) appreciates this opportunity to submit comments in response to the National Science Foundation Networking and Information Technology Research and Development (NITRD) National Coordination Office's (NCO) Request for Information. As a leading provider of digital transformation solutions to more than 99% of the Fortune 500 and 15 cabinet-level Federal agencies, Deloitte provides insights, implementation, and operational support that deliver on the promise of Artificial Intelligence ("AI") innovation.

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Our comments reflect our deep experience with more than 2,800 customers who are applying leading-edge artificial intelligence processes across industry and government. We look forward to working with you to achieve optimum value to the National Coordination Office. If you have any questions or require additional information, please contact me at +(703) [redacted] or Roger Sion at + (571) [redacted].

Sincerely,

Edward Van Buren
Principal
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Roger Sion
Managing Director
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[1] As used in this document, "Deloitte" means Deloitte Consulting, LLP, a subsidiary of Deloitte LLP. Please see www.deloitte.com/us/about for a detailed description of the legal structure of Deloitte LLP and its subsidiaries. Certain services may not be available to attest clients under the rules and regulations of public accounting.

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Introduction

Deloitte applauds NITRD NCO's efforts to develop an AI Action Plan to implement the President's Executive Order on Removing Barriers to American Leadership in Artificial Intelligence (Executive Order 14179).

U.S. led commercial innovation has led to the development of transformative AI capabilities from automating repetitive tasks to Generative AI (GenAI) solutions that can use reasoning to answer questions that drive mission outcomes and government efficiency. Today, companies stand on the verge of breakthroughs in solutions, like agentic AI, which are poised to accelerate human performance and enable AI at scale. Far from being an abstract competition, AI breakthroughs are essential to enhancing the economic competitiveness, national security, and wellbeing of everyday Americans. Similar to previous eras of American-led technology advancements (from railroads to the internet), advancements in AI are poised catalyze innovations in diverse fields, including quantum computing, cryptocurrencies, robotics, and nanotechnology.

However, as AI innovation continues to expand, America faces a more complex competitive environment with a number of dimensions. AI leadership requires the development of superior AI technologies, faster adoption of those systems by users, and enduring business models that accelerate AI transformation. Each of these dimensions of competition is integral to continued U.S. leadership in AI.

- **Competition for technology innovation** determines U.S. ability to maintain its dominant position in AI and will help American businesses and workers.[i]
- **Competition for adoption** determines how quickly Americans will reap the benefits of AI through greater per capita GDP and stronger national security.
- **Competition on business models** determines which models and practices dominate the global AI industry and the extent to which American business models shape the AI landscape.[ii]

To sustain and grow its lead in AI, America needs government, industry, academia, and other players to work together across each of these dimensions of AI competition. This cooperation must be done efficiently, not be tied down by bureaucracy. With decades of working with government, industry, and academia on AI, Deloitte is keenly interested in supporting American AI development. Deloitte is a leading digital transformation firm in the field of AI, with 27,000+ AI skilled practitioners efficiently working across government and private sector companies. We leverage software innovation from our leading technology partners to drive efficiencies and mission outcomes.

Observations and Considerations

To support economic competitiveness, national security, and human flourishing through innovation, **the United States needs to win** in each dimension of AI competition. To this end, we recommend the government consider the following actions:

Win the Technology Race

- **Adopt hybrid cloud infrastructure to accelerate AI solution development**: Given its scale, government can act as a catalyst for faster and scalable AI infrastructure development. With a majority of AI workloads being hosted in public, private, or hybrid cloud architecture, the U.S. government should ensure that it builds the infrastructure required to deploy both current AI solutions and frontier models that are pushing the limits of AI capabilities.[iii] This requires stronger partnerships with startups, innovation labs, and academic research centers to inform infrastructure requirements.
- **Provide risk management guidance to safeguard future growth**: The AI Action Plan can help to secure AI by creating voluntary standards and guidance for AI risk mitigation so that that AI is used responsibly. Favoring outcomes-based voluntary standards will enable AI innovations that outpace our competitors and shape global AI standards in America's best interest.

Win the Adoption Race

- **Accelerate AI innovation by streamlining procurement processes**: Accelerating government adoption can improve government efficiency, accelerate the delivery of government services to Americans, help AI startups to grow more quickly and spur broader adoption of AI across the country. Techniques like Other Transaction Authorities (OTAs) and expanding the Simplified Acquisition Threshold can speed government's adoption.
- **Streamline and expedite approvals for cutting-edge AI technologies:** The Federal government can help to accelerate adoption of cutting-edge technology by streamlining duplicative information reviews present in existing information security processes, such as the Federal Risk and Authorization Management Program (FedRAMP) and Authority to Operate (ATO) processes. This would improve efficiency while maintaining a strong security posture.
- **Right access to the right workers with the right skills:** With robust support from senior leadership, line employees in government stand poised to unearth many of the most transformative AI use cases.[iv] But to find and develop those winning use cases, workers need access to the appropriate tools and level of training for their roles and occupations.

Win the Enduring Business Model Race

- **Promote AI innovation through leveraging open-source development standards:** The National Institute of Standards and Technology (NIST) has the opportunity to accelerate American-led AI innovation and faster AI adoption by strengthening open-source AI standards in its Risk Management Framework.

- **Support competitive business relationships:** America's private sector is already one of the most trusted globally. The Government should encourage cross-sector cooperation and partnerships domestically to increase innovation and support American AI growth in the global market.

Win the Technology Race

Adopt hybrid cloud infrastructure to accelerate AI solution development

Given the vast size and scale of information processing of the government, there is an opportunity for the government to act as a catalyst for faster and scalable infrastructure development.

Widespread AI adoption requires stronger infrastructure. For government, this starts with improved cloud technology. While AI models are shrinking and significant AI advancements are making on-device models a reality, the vast majority of government AI workloads are still hosted in some form of cloud environment. This means that building sufficient cloud infrastructure – both physical data centers as well as software and managed services – across the country is central to continued AI strength.

Significant infrastructure is needed to support the rapid pace of AI and cloud adoption. That means physical infrastructure such as power and water for data centers. Deloitte predicts data centers will only make up about 2% of global electricity consumption, or 536 terawatt-hours (TWh), in 2025. But as power-intensive GenAI training and inference continues to grow faster than other uses and applications, global data center electricity consumption could roughly double to 1,065 TWh by 2030. [v] Along with the power, the U.S. will also need the talent and software that go into the cloud-based managed services.

Federal buying power can help move the needle on both fronts. The integration of AI into government's existing digital infrastructure remains a significant challenge, both in terms of financial investment and speed of deployment. To deploy AI applications, and frontier models, agencies require high bandwidth, low latency and flexible architectures to train and fine-tune AI applications and deploy them to a global workforce that operates on-prem, on cloud, on-edge, and on device. This necessitates substantial financial investment and the ability to scale rapidly, which can be particularly daunting for government agencies operating in the era of rapid technological innovation.

The challenges associated with building AI infrastructure can lead to several implications for government agencies, including delayed implementations, increased costs, technological lag, and operational inefficiencies. For instance, the lack of a robust AI strategy could result in fragmented efforts and missed opportunities for innovation. Moreover, without proper data governance practices, agencies may face potential cybersecurity and national security challenges, as highlighted by the 2030 Vision of the Federal Data Strategy[vi].

To overcome these challenges, government can play a pivotal role by fostering an environment where emerging AI technologies can thrive. This involves ensuring diverse perspectives are heard in public-private partnerships, establishing data governance practices, and promoting secure and reliable AI.

Provide risk management guidance to safeguard future growth

The AI Action Plan can advance AI innovation by creating voluntary consensus standards and guidance for AI risk mitigation to ensure that the technology is used responsibly.

AI holds great promise for promoting U.S. economic growth, national security, and technological competitiveness. AI can also introduce new risks if deployed without sufficient human oversight and risk management processes. AI can act against the goals of its creators and, in more extreme adversarial scenarios, could cause damage to U.S. national and economic security.

Uncertainty about how these risks will be controlled – whether by regulation or by other means – impairs American innovation.[vii] America's future strength in AI rests on finding clarity.

More than simply human oversight, AI development needs American oversight. Various countries, many of which are America's greatest geopolitical competitors, already use AI as an autocratic tool.[viii]

A clear focus on desirable and undesirable outcomes can help government protect public interests that will help industry to continue innovating. Despite being a powerful tool, outcome-based policies – which describe the intended result that a regulation should deliver, rather than describing the specific sequences of actions – make up less than 1% of the 1,600+ AI policies enacted across the globe.[ix]

Voluntary consensus standards and guidance for AI risk mitigation can help turn that focus on outcomes into actions that organizations can take. NIST has played an important role over the years in developing voluntary consensus standards such as the NIST AI Risk Management Framework (AI RMF), Cybersecurity Framework, and Privacy Risk Management Framework. NIST's long track record of creating flexible, outcome-based, and adoptable risk management frameworks that are voluntary but widely adopted by industry can be incorporated into the new plan or serve as a model for new standards both nationally and globally.

NIST's deep AI experience, which began during the first Trump Administration at the direction of the President's 2019 Executive Order (Executive Order 13859), positions the agency well to play a powerful role in the AI Action Plan. When NIST was developing the AI RMF, Deloitte recommended that creating a voluntary, flexible, outcome based, and highly adoptable framework is the correct approach to create "guardrails" for organizations use to engender a trustworthy AI marketplace that can help enable innovation while mitigating harm. These goals remain relevant as the U.S. develops an AI Action Plan. In addition, the AI RMF can serve as an example as an iterative document. Because AI moves at a rapid pace, it can be beneficial for frameworks to adapt to new findings and input from industry, academia, civil society organizations, and other stakeholders.

Win the Adoption Race

Accelerate AI innovation through streamlining procurement processes

Government procurement can support American-led AI innovation by expanding the use of rapid contracting techniques like OTA and expanding the Simplified Acquisition Threshold.

As a leading buyer of information technology, the Federal government can shape the global AI landscape by supporting the American-led AI ecosystem of startups and technology innovators through procurement. This is a role that the Federal government has played in the past. For example, as one of the largest single buyers of cloud services, Federal performance standards helped promote cloud security practices across the industry.[x] To drive AI innovation, the Federal government can follow this pattern and use its scale to apply innovations from private industry. This would both accelerate government efficiency and strengthen the American AI innovation, especially among startups.

However, complex procurement regulations and outdated requirements present a significant barrier to bringing AI innovation into government. It takes agencies an average of 18-24 months to award contracts, with further delays for the acquisition of innovative technologies like AI. Moreover, government contracting continues to rely on overly prescriptive requirements that often rely on outdated technologies, manual solutions, and business-as-usual thinking. These factors can result in the delivery of outdated solutions by the time contracts are awarded, resulting in inefficient technology that is in some cases decades behind industry standards.

Consequently, startup firms – where many AI innovations are pioneered - are discouraged from participating in government contracts due to the lengthy and cumbersome procurement processes. This not only stifles the ability of innovators to drive government efficiency but also misses an opportunity to further extend America's AI leadership in the commercial market.

The government has several options to streamline procurement to support AI innovation. Leveraging alternative contracting mechanisms such as OTAs have improved procurement efficiency by streamlining reviews. OTAs allow for more flexible and expedited procurement processes, reducing the award timelines to as little as 3 to 6 months. This makes it particularly useful for acquiring innovative products at a lower risk to the government.

Expanding the use of Simplified Acquisition Thresholds can also improve procurement timelines and enable the faster adoption of innovative AI solutions at low cost and risk to the government. Because Simplified Acquisition Threshold acquisitions generally require a lower administrative burden, they can more effectively be managed by procurement teams.

Streamline and expedite approvals for cutting-edge AI technologies

The Federal government can accelerate adoption of cutting-edge technology by streamlining technology approvals, such as FedRAMP and agency specific ATOs, to improve efficiency while maintaining a strong security posture.

As stewards of taxpayer dollars, the government has a responsibility to efficiently and effectively fulfill its mission obligations, and advances in AI technology present a significant opportunity to improve efficiency and government effectiveness. However, the AI technology ecosystem is rapidly evolving, and agencies risk being stymied by their inability to keep pace with this acceleration. Faster technology approvals will enable agencies to more readily adopt and scale cutting-edge technologies, resulting in continued cost savings and efficiency gains.

Without cost effective, timely, and accessible technology approvals, Federal agencies may miss out on applying advancements like agentic AI to transform government. For example, while the FedRAMP is an important process for standardizing cloud security across agencies, the process can be burdensome, which prevents agencies and Cloud Service Providers (CSPs) from pursuing authorization. Common barriers to wider FedRAMP adoption include the cost and complexity for CSPs and agencies coupled with limited staff to oversee the authorization process.

While efforts have been made in recent years to shorten the timeline and complexity, the AI Action Plan could include additional considerations to streamline and speed up the approval process without compromising security standards:

- Regularly review FedRAMP standards and employ a more tailored set of baseline FedRAMP security requirements to focus on the most pressing threats and eliminate excess controls
- Employ automation and agentic AI where possible to streamline reviews and deliver better continuous monitoring
- Provide opportunities for re-use of documentation or assessments between FedRAMP and ATO where appropriate
- Allow use of AI sandboxes to test pre-authorized tools prior to deployment in a controlled production environment
- Increase acceptance of pre-approved external certifications

The right access to the right workers with the right skills

The best technology is of little value if people do not use it. Providing government workers the specific skills and fluency they need to use AI in their unique role can help spur adoption and find hidden AI innovations.

Unlike commercial enterprises in which leadership most often drives AI scaling, line employees in government stand poised to unearth many of the most transformative AI use cases. But to find and develop those winning use cases, workers need access to the appropriate tools and level of training for their roles and occupations. Access to GenAI continues to be an issue, with only 1% of government respondents in a recent Deloitte survey reporting that 60% of their workforce had access to GenAI tools.[xi] This is beginning to change, albeit slowly, as more governments give their workers wider access to GenAI tools.

While government access to tools continues to grow slowly, leaders can move to address the AI fluency issue immediately. Not every worker needs to know how to fine-tune a large language model or build their own chatbot. While a few users will need detailed knowledge on how to build AI tools, others will benefit from mid-level knowledge on how to select tools while others need only basic knowledge on how to use tools. Our research suggests that this build-choose-use paradigm for AI fluency varies with a worker's occupation, level, and role. Occupations with higher exposure to GenAI—and higher *potential* exposure—need more knowledge, to best take advantage of the technology's availability. Similarly, managers likely need more AI fluency than entry-level workers who may just need to know how to use tools in a finite number of situations. Of course, those whose role involves creating AI tools to enhance current processes and potential new ones will need more skill than those in technical- or end-user roles.

These tiers of AI fluency can help the workforce quickly gain the right level of AI knowledge for them. This can be important when setting expectations for what the experience of working with AI will be like. With so much hyperbole floating around, expectations can be high for near magical experiences, and one bad experience with AI can stall further experimentation for a whole organization. So, making progress on AI depends on having the right fluency – both skills and expectations – for every individual.

With the right access to tools and the right skills, government workers can find new opportunities for AI to benefit the public at an unprecedented pace and scale.

Win the Enduring Business Model Race

Promote AI innovation through leveraging open-source development standards

The race for technology and adoption is only viable if companies have business models that can endure. With adversaries racing to attract as many users as possible, NIST has an opportunity to accelerate American-led AI innovation by prioritizing open-source AI standards.

In recent years, several open-source AI model weights have been released, including X.ai's Grok and Google's Gemma models. These models enable researchers to read, contribute to, and improve the models at the heart of AI innovation. By promoting transparency, collaboration, and accessibility, open-source AI enables government agencies, private sector companies, and academic researchers to innovate and build from each other's advancements.

Open-source approaches have also challenged U.S. advantages in technology and AI adoption. Because open-source AI solutions generally require lower capital investment, they can garner users more quickly than closed-source models. That advantage has traditionally been offset by more modest performance of open-source models, but recent AI advancements are erasing those performance gaps. This presents a challenge to U.S. firms, which have not only conducted the most valuable research on AI but also followed traditional closed-source R&D product development approaches.

For NIST, affirming open-source standards to develop AI solutions would deliver several benefits. First, it would improve innovation in the domestic AI marketplace. Because researchers could build on each other's advancements, open-source AI would reduce barriers to market entry and enable new product and service innovations. Second, by specifying what components of AI solutions should be based on an open standard while protecting underlying AI model weights, NIST could protect intellectual property and manage AI risk. Third, by lowering capital investment required to develop AI solutions, it could expand the global reach of U.S. models and solutions.

Support competitive business models

Government can support the creation of stable, enduring business models by promoting the sharing of ideas across industries.

American AI companies and technologies could benefit from practices that promote a competitive and open marketplace while ensuring a level playing field. For example, encouraging deep collaboration between government, industry, and academia can lead to shared best practices in research, development, and intellectual property protection. Building a framework that values innovation and transparency can help companies confidently invest in new ideas and expand their operations.

Moreover, fostering reciprocal business relationships and embracing strategic partnerships can open new opportunities for advancing AI products and services. Emphasizing shared responsibility and mutual benefit can lead to practices that support sustainable growth while safeguarding national economic interests. By creating environments where innovation thrives and competitive advantages are recognized fairly, American AI firms may naturally rise to prominence and make a positive impact on the global market.

Conclusion

The effort undertaken by OSTP to create an AI Action Plan represents a distinct opportunity to chart a path for the United States to sustain and enhance America's AI leadership in order to promote human flourishing, economic competitiveness, and national security. Deloitte leaders in AI across multiple sectors would be pleased to continue the conversation and share our knowledge from experiences working to implement AI strategies on behalf of government and commercial clients.

[i] Richard Fontaine and Kara Frederick, "The Autocrat's New Tool Kit," The Wall Street Journal, March 15, 2019.
[ii] Bruce Chew. "DeepSeek: The markets and the industry," Purdue University Tech Diplomacy Academy, February 2025.
[iii] "Weighing the Open-Source Hybrid Option for Adopting Generative AI." Harvard Business Review Briefing Paper. 2023.
[iv] Joe Mariani, Pankaj Kishnani, Ahmed Alibage, "Government's less trodden path to scaling generative AI," Deloitte Insights, October 24, 2024.
[v] Karthik Ramachandran, Duncan Stewart, Kate Hardin, Gillian Crossan. "As generative AI asks for more power, data centers seek more reliable, cleaner energy solutions." Deloitte Insights. November 19, 2024.
[vi] Chief Data Officer Council. "Federal data strategy: leveraging data as a strategic asset." Office of Management and Budget, 2021
[vii] Joe Mariani, William Eggers, Pankaj Kishnani. "The AI regulations that aren't being talked about: Patterns in AI policies can expose new opportunities for governments to steer AI's development." Deloitte Insights. November 10, 2023.
[viii] Richard Fontaine and Kara Frederick, "The Autocrat's New Tool Kit," The Wall Street Journal, March 15, 2019.
[ix] Joe Mariani, William Eggers, Pankaj Kishnani, "The AI regulations that aren't being talked about: Patterns in AI policies can expose new opportunities for governments to steer AI's development," Deloitte Insights. November 10, 2023.
[x] According to the annual FedRamp survey of government and industry professionals, 85% agreed that FedRAMP promotes the adoption of secure cloud services across the U.S. Government. FedRAMP, "FedRAMP FY22 annual survey recap," January 17, 2023.
[xi] Joe Mariani, Pankaj Kishnani, Ahmed Alibage, "Government's less trodden path to scaling generative AI," Deloitte Insights, October 24, 2024.